# Week 03: Literature Review and Bibliometrics

## R&D and Project Management in Computer Science

R&D and Project Management in Computer Science (11117BLG002). Prof. Dr. Utku Kose. Full Professor, Department of Computer Engineering, Süleyman Demirel University, Isparta, Türkiye. Additional affiliations: University of North Dakota (USA), Universidad Panamericana (Mexico City, Mexico), Vel Tech University (Chennai, India). ORCID 0000-0002-9652-6415. E-mail: utkukose@sdu.edu.tr, utku.kose@und.edu, ukose@up.edu.mx, utkukose@gmail.com. This course is updated in line with current developments in the field. First version: September 2026. Last update: October 2026.

## Overview

Originality can only be claimed against the literature. This week covers systematic reviews and mapping studies in computing, open bibliographic data and bibliometric indicators, together with the principles that limit their use [1, 2, 3].

## Learning outcomes

By the end of the week, students are expected to plan a systematic literature review or mapping study and to report its selection process with a flow diagram. They should also be able to retrieve publication data from OpenAlex, and to compute and critically interpret indicators such as the h-index.

## Lecture

### Systematic reviews in computing

Kitchenham and Charters adapted systematic review methods from medicine to software engineering [1]. A review is planned in a protocol, so that others can repeat it. The protocol states the research questions, the search strategy, the inclusion and exclusion criteria, the quality assessment and the synthesis method. Petersen, Vakkalanka and Kuzniarz updated guidelines for systematic mapping studies, which classify a research field rather than answer a narrow question [4]. Wohlin described snowballing, which follows the references of relevant papers backwards and their citations forwards, as a complement or alternative to database searches [5].

### Reporting the selection

The PRISMA 2020 statement provides a reporting guideline for systematic reviews, with a checklist and a flow diagram that accounts for every record from identification to inclusion [6]. Although it was developed for health research, its flow diagram is widely used in computing. Figure 3.1 shows the structure with illustrative numbers. The interactive lab checks the arithmetic of such a flow, since inconsistent numbers are a common and easily avoided error.

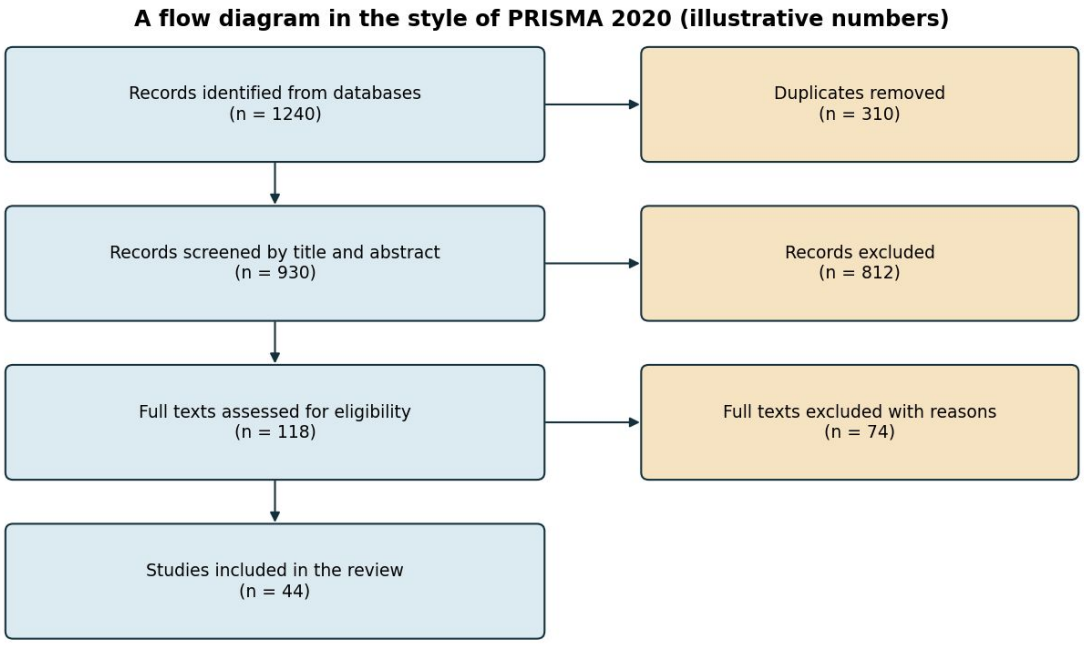



*Figure 3.1. The selection process of a systematic review in the style of the PRISMA 2020 flow diagram, with illustrative numbers [6].*

## Open bibliographic data

OpenAlex is a fully open index of scholarly works, authors, sources, institutions and concepts, available through a free programming interface [2]. It allows students to count publications on a topic per year, to find the most cited works and to build datasets for mapping studies without subscriptions. Tools such as VOSviewer and bibliometrix support the visualisation and analysis of such data [7, 8]. The interactive lab queries OpenAlex live, and the Colab notebook retrieves the same data with Python, falling back to a small illustrative sample when no connection is available.

## Indicators and their limits

Hirsch proposed the h-index: A researcher has index h if h of their papers have at least h citations each [9]. The index is easy to compute and to misuse. It depends on field, career length and database coverage. The Leiden Manifesto set out ten principles for research metrics, among them that quantitative evaluation should support qualitative expert assessment rather than replace it [3]. The San Francisco Declaration on Research Assessment recommends against using journal-based metrics, such as journal impact factors, as a surrogate measure of the quality of individual articles in funding, hiring and promotion decisions [10]. For a proposal, bibliometrics serve best as evidence about the state of a field, such as growth and gaps, not as a verdict on quality.

> ***Pause and reflect.*** *What would a mapping study of your topic reveal that a narrative review would not?*

## Hands-on components

Interactive lab: Part A queries the open OpenAlex index for publication counts per year and the most cited works on a topic. Part B computes the h-index and related indicators from a list of citation counts. Part C checks the arithmetic of a review flow diagram [2, 6, 9]. Link: https://utkukose.github.io/rd-project-management-NB-lecture/weeks/week-03/lab.html

Colab notebook: It retrieves publication counts from the OpenAlex interface, with an illustrative fallback sample when offline, computes growth and implements the h-index [2, 9]. Link: https://colab.research.google.com/github/utkukose/rd-project-management-NB-lecture/blob/main/weeks/week-03/NB03_literature_bibliometrics.ipynb

## Reflection prompts

1. What did the OpenAlex query show about your topic, and how would you use that evidence in a proposal?
2. Which of the ten Leiden principles is most often ignored in your environment?
3. Draft the inclusion and exclusion criteria for a review of your topic. Which would be hardest to apply consistently?

## Weekly task and submission

**What to do.** Write a review protocol for the topic of your project, about 500 words. It states the research questions of the review, the databases and the search string, the inclusion and exclusion criteria and the snowballing plan. Then run a first search and draw a flow diagram with its numbers. Add one chart of publication counts per year from OpenAlex.

**Where to look.** Lecture: sections "Systematic reviews in computing", "Reporting the selection" and "Open bibliographic data". Interactive lab: Part A for a live OpenAlex query and Part C for checking the arithmetic of a flow diagram. Colab notebook: Section 1, which retrieves publication counts from OpenAlex.

**Example.** For a review of energy-efficient training of language models, the search string could be ("energy" OR "carbon") AND "training" AND "language model". One inclusion criterion could require a measured energy or carbon figure. The flow diagram could report 412 records found, 120 records left after screening titles and abstracts, and 35 studies included.

**What to submit.** The protocol with the flow diagram and the OpenAlex chart, and the Colab notebook with Exercises 1 and 2 completed.

The weekly task supports self-learning and builds a personal portfolio. In an active semester, students who follow the course with the instructor can send their work for evaluation to utkukose@sdu.edu.tr or utkukose@gmail.com. The e-mail contains the items that the task lists under What to submit. It also includes the learning log exported from the interactive lab (tab: Reflect and submit).

## References

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